

LWE and Ring-LWE

The Number-Theoretic Structure of Ideal-Lattice Cryptography — Final Oral Report (~40 min)

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Geometry of numbers $\rightarrow$ cyclotomic fields & ideal lattices $\rightarrow$ **LWE / Ring-LWE** $\rightarrow$
post-quantum standards

Part 0 — Preliminaries [~ 10 min]

- Lattice
- Minkowski's first theorem
- Cyclotomic field
- Ideal-lattice structure
- Shortest Vector Problem (SVP)
- Closest Vector Problem (CVP)

Part 1 — LWE (core) [~ 12 min]

- Setup; decision vs. search versions
- Lattice connection; security reduction
- Hardness and efficiency bottleneck

Part 2 — Ring-LWE (core) [~ 13 min]

- Cyclotomic-ring setup; prime-ideal splitting
- Fast NTT multiplication; dual-lattice error
- Sample construction and reduction chain

Part 3 — Applications & open problems [~ 5 min]

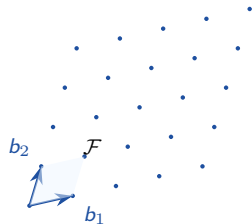
Definition

Take n linearly independent vectors $b_1, \dots, b_n$ in $\mathbb{R}^n$; their integer linear combinations

$$\mathcal{L} = \sum_{i=1}^n \mathbb{Z} b_i \subset \mathbb{R}^n$$

form a **lattice** of rank n . The quantity $\det(\mathcal{L}) := |\det B|$ of the basis matrix $B = [b_1 | \dots | b_n]$ is the **covolume**, independent of the chosen basis, and geometrically equals the volume of the fundamental domain $\mathcal{F} = B[0, 1)^n$.

- One lattice has infinitely many bases; the gap between a “good” basis (short, nearly orthogonal) and a “bad” one is the source of hard problems.
- The smaller the covolume, the denser the lattice points.
- Next: an ideal in a number field $\mapsto$ a lattice (ideal



Theorem

Let $\mathcal{L} \subset \mathbb{R}^n$ be a lattice and $S \subset \mathbb{R}^n$ a **centrally symmetric convex body** ($x \in S \Rightarrow -x \in S$; bounded, closed, convex). If

$$\text{vol}(S) > 2^n \det(\mathcal{L}),$$

then S contains at least one nonzero lattice point $v \in \mathcal{L} \setminus \{0\}$.

- **Proof idea:** apply Blichfeldt's pigeonhole argument to $\frac{1}{2}S$; $\text{vol}(\frac{1}{2}S) > \det(\mathcal{L})$ forces two translated copies to overlap, then central symmetry + convexity pull the difference vector back into S .
- **Key message:** “large enough volume $\Rightarrow$ a nonzero lattice point must exist” — a **non-constructive** existence theorem: it guarantees a short vector *exists* but gives no way to find it.
- This is precisely the geometric root of the hardness of SVP and the foundation of lattice-based security.

Setup (take $n = 2^k$, power-of-two cyclotomic ring)

$\zeta = \zeta_{2n} = e^{i\pi/n}$ is a primitive $2n$ -th root of unity; the cyclotomic polynomial is $\Phi_{2n}(x) = x^n + 1$ (irreducible over $\mathbb{Q}$ when $n = 2^k$), $K = \mathbb{Q}(\zeta)$, $[K : \mathbb{Q}] = n$.

- The ring of integers $\mathcal{O}_K = \mathbb{Z}[\zeta] \cong \mathbb{Z}[x]/(x^n + 1)$ is a **Dedekind domain** (Noetherian, integrally closed, every nonzero prime ideal maximal) $\Rightarrow$ unique factorization of every nonzero ideal into prime ideals.
- The **canonical embedding** $\sigma_j(\zeta) = \zeta^{2^{j-1}}$ ($j = 1, \dots, n$) gives $\sigma : K \hookrightarrow \mathbb{C}^n$; conjugate symmetry places the image in $H \cong \mathbb{R}^n$, and it is compatible with ring multiplication (component-wise).
- Discriminant: $|\Delta_K| = (2n)^n / 2^n = n^n$.
- Why $x^n + 1$: cleanest irreducibility; $x^n \equiv -1$ gives “negacyclic convolution”; friendly to the NTT.

Ideal $\xrightarrow{\sigma}$ lattice

A fractional ideal $\mathfrak{a} \subset \mathcal{O}_K$ maps under the canonical embedding to $\sigma(\mathfrak{a}) \subset H$, a rank- n lattice closed under multiplication by $\mathcal{O}_K$ (hence an **ideal lattice**).

- Exact relation between covolume and ideal norm $N(\mathfrak{a}) = [\mathcal{O}_K : \mathfrak{a}]$:

$$\det(\sigma(\mathcal{O}_K)) = \sqrt{|\Delta_K|} = n^{n/2}, \quad \det(\sigma(\mathfrak{a})) = N(\mathfrak{a}) \cdot n^{n/2}.$$

- **Meaning:** it binds the (arithmetic) “ideal norm” to the (geometric) “lattice volume”; ideal multiplication $\leftrightarrow$ an algebraic action on the lattice.
- An ideal lattice is a special case of a general lattice, but carries extra ring structure $\Rightarrow$ **the source of efficiency** (Ring-LWE) and also **the source of security debate** (is ideal-SVP easier?).

Problem

Given a (bad) basis of a lattice $\mathcal{L}$, find the length of the shortest nonzero vector $\lambda_1(\mathcal{L}) = \min_{v \in \mathcal{L} \setminus \{0\}} \|v\|$, or such a v .

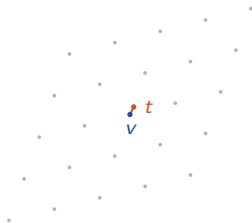
- **Approximate SVP** $_{\gamma}$: find $v \neq 0$ with $\|v\| \leq \gamma \cdot \lambda_1$.
- NP-hard (under randomized reductions, $\gamma = O(1)$); best algorithms run in $2^{\Theta(n)}$.
- Best quantum algorithm $2^{0.265n}$ (Laarhoven 2015) $\Rightarrow$ still hard post-quantum.
- The security of LWE/Ring-LWE ultimately reduces to this (worst case).



Problem

Given a lattice $\mathcal{L}$ and a target point $t \in \mathbb{R}^n$ (not necessarily on the lattice), find the nearest lattice point $v \in \mathcal{L}$ minimizing $\|t - v\|$.

- Generally harder than SVP ($\text{SVP} \leq \text{CVP}$ by reduction).
- **BDD (Bounded Distance Decoding)**: a special case of CVP where t is known to lie very close to some lattice point ($< \lambda_1/2$); recover it $\Rightarrow$ unique solution.
- **LWE is essentially BDD**: in $b = As + e$, As is a lattice point and e a small error, so recovering s = solving BDD on a q -ary lattice.



find the lattice point nearest to t

Learning With Errors

Fix a dimension n , modulus q , and error distribution χ (discrete Gaussian of width αq , $\alpha \ll 1$). Secret $s \in \mathbb{Z}_q^n$. Each sample:

$$a \xleftarrow{\$} \mathbb{Z}_q^n, \quad b = \langle a, s \rangle + e \bmod q, \quad e \leftarrow \chi.$$

- Collecting m samples in matrix form, $A \in \mathbb{Z}_q^{m \times n}$,

$$b = As + e \bmod q \in \mathbb{Z}_q^m.$$

- Intuition: *without* the error e , n linear equations and Gaussian elimination recover s ; **adding a tiny error breaks linear algebra** — this is where the hardness lies.
- The error e must be “small enough for correctness, large enough for security,” controlled by α .

Search-LWE

Given many samples (a_i, b_i) , **recover the secret s** .

Decision-LWE

Distinguish the two distributions:

$(a, \langle a, s \rangle + e)$ vs. (a, u) , with $u \xleftarrow{\$} \mathbb{Z}_q$ uniform.

- The two are **polynomial-time equivalent** (search-to-decision; Regev's reduction for prime q).
- **Encryption builds on the decision version**: ciphertexts indistinguishable from random $\Rightarrow$ semantic security (IND-CPA).
- Example (Regev encryption): public key $(A, b = As + e)$; to encrypt a bit μ , take a subset sum $(\sum a_i, \sum b_i + \mu \lfloor q/2 \rfloor)$; decrypt by checking whether the inner product is near 0 or $q/2$.

BDD on a q -ary lattice

Let $\Lambda_q(A) = \{y \in \mathbb{Z}^m : y \equiv As \bmod q, \exists s \in \mathbb{Z}_q^n\}$. Then $b = As + e$ is a **target close to the lattice point** As , with error e as the offset $\Rightarrow$ recovering s = solving **BDD/CVP** on $\Lambda_q(A)$.

Regev's quantum reduction (worst-case $\rightarrow$ average-case)

worst-case $\text{GapSVP}_\gamma / \text{SIVP}_\gamma \xrightarrow{\text{quantum poly-time}} \text{average-case LWE}, \quad \gamma = \tilde{O}(n/\alpha).$

- Meaning: **breaking a random LWE instance** $\Rightarrow$ **solving a worst-case lattice problem** — security rests on the “hardest” instances.
- Peikert (2009) later added a classical reduction (for GapSVP , large q).

- **Pros:** a clean security reduction with no extra algebraic assumptions $\Rightarrow$ the most conservative lattice foundation.
- **Fatal bottleneck (efficiency):**
 - The public key is an $m \times n$ matrix $A \Rightarrow$ size $O(n^2)$ (indeed $\tilde{O}(n^2)$).
 - Each operation is a generic matrix–vector product $\Rightarrow O(n^2)$ time.
 - For practical security ($n \sim 10^3$), keys reach several MB — hard to deploy.

The central question

Can we cut $O(n^2)$ down to nearly linear **without sacrificing the security reduction?**

Solution: give the secret and samples a “ring structure” $\Rightarrow$ one ring element $a \in R_q$ yields n equations at once; key $O(n)$, multiplication $O(n \log n)$. This is **Ring-LWE**.

Ring setup

$R = \mathcal{O}_K = \mathbb{Z}[x]/(x^n + 1)$ ($n = 2^k$), $R_q = R/qR \cong \mathbb{Z}_q[x]/(x^n + 1)$. Secret $s \in R_q$, error $e \leftarrow \chi_R$ (narrow discrete Gaussian on R). Sample:

$$a \xleftarrow{\$} R_q, \quad b = a \cdot s + e \in R_q.$$

- Compared with LWE: the vector space $\mathbb{Z}_q^n \rightarrow$ ring R_q ; inner product $\rightarrow$ **ring multiplication** (negacyclic convolution).
- A single $a \in R_q$ (n coefficients) carries n “equations” $\Rightarrow$ key drops from $O(n^2)$ to $O(n)$.
- Security reduces to SVP on **ideal lattices** (rather than arbitrary lattices).
- The algebraic structure (prime-ideal splitting, CRT, dual lattice) enables fast computation and a valid reduction.

For a prime $q \nmid 2n$, the splitting of q in $\mathcal{O}_K$ is determined by the factorization of $\Phi_{2n}(x) \bmod q$:

$$\Phi_{2n}(x) \equiv \prod_{i=1}^g \phi_i(x)^{e_i} \pmod{q}, \quad \deg \phi_i = f, \quad efg = n, \quad q\mathcal{O}_K = \mathfrak{q}_1^{e_1} \cdots \mathfrak{q}_g^{e_g}.$$

Complete-splitting condition

q splits completely ($e = f = 1$, $g = n$) $\iff q \equiv 1 \pmod{2n}$. Then $\mathbb{F}_q^\times$ contains a primitive $2n$ -th root of unity ω , with $\Phi_{2n}(x) \equiv \prod_{j=1}^n (x - \omega_j) \pmod{q}$, so by CRT:

$$R_q \xrightarrow{\cong} \prod_{j=1}^n \mathbb{F}_q[x]/(x - \omega_j) \cong \mathbb{F}_q^n, \quad a(x) \mapsto (a(\omega_1), \dots, a(\omega_n)).$$

- This ring isomorphism turns **polynomial multiplication** into **component-wise multiplication** in $\mathbb{F}_q^n$ — the algebraic root of the NTT.

Number-Theoretic Transform (NTT)

Under complete splitting, define the evaluation map

$\hat{a} = \text{NTT}(a) := (a(\omega_1), \dots, a(\omega_n)) \in \mathbb{F}_q^n$. Then

$$\text{NTT}(a \cdot b) = \text{NTT}(a) \odot \text{NTT}(b) = (a(\omega_j) b(\omega_j))_{j=1}^n,$$

after which the inverse transform NTT^{-1} (interpolation, with $n^{-1} \bmod q$ scaling) recovers the coefficients.

- **Cooley–Tukey butterfly**: recursive split by parity of indices, $a(\omega_j) = a_{\text{even}}(\omega_j^2) + \omega_j a_{\text{odd}}(\omega_j^2)$; with $\omega^n = -1$, $T(n) = 2T(n/2) + O(n) = O(n \log n)$.
- $x^n + 1 \Rightarrow x^n \equiv -1$, so ring multiplication is **negacyclic convolution** (wrap-around terms flip sign) — power-of-two cyclotomic rings keep this cleanest.
- Efficiency: direct polynomial multiplication $O(n^2) \rightarrow \text{NTT } O(n \log n)$.

Different and codifferent (dual lattice)

Different ideal $\mathfrak{D}_{K/\mathbb{Q}} = (\Phi'_{2n}(\zeta)) = (n\zeta^{n-1}) = (n)$; under the trace pairing $\langle a, b \rangle = \text{Tr}_{K/\mathbb{Q}}(ab)$, the **dual lattice** of $\mathcal{O}_K$ is the codifferent

$$\mathfrak{D}^{-1} = \{x \in K : \text{Tr}_{K/\mathbb{Q}}(x\mathcal{O}_K) \subset \mathbb{Z}\} = \frac{1}{n}\mathcal{O}_K.$$

- Key to the verification: $\text{Tr}_{K/\mathbb{Q}}(\zeta^j) = n$ (for $j = 0$) or 0 (for $1 \leq j \leq n-1$).
- **Cryptographic meaning:** in rigorous Ring-LWE the error lives in $\mathfrak{D}^{-1}/q\mathfrak{D}^{-1} = \frac{1}{n}R_q$ (not R_q itself); only the dual lattice keeps the error distribution **algebraically invariant** under the trace pairing, ensuring a tight reduction.
- In power-of-two cyclotomic rings, since $\mathfrak{D} = (n)$ is principal, this correction is just a $\frac{1}{n}$ scaling — simple to implement.

Security reduction chain (quantum, LPR 2010)

worst-case ideal-SVP _{γ} $\xrightarrow{\text{quantum poly}}$ average-case Ring-LWE $\Rightarrow$ breaking Kyber/Dilithium is

	Key size	Mult. cost	Security basis
LWE (Regev 2005)	$O(n^2)$	$O(n^2)$ matrix mult.	arbitrary-lattice SVP/SIVP
Ring-LWE (LPR 2010)	$O(n)$	$O(n \log n)$ NTT	ideal-lattice ideal-SVP

- The algebraic toolkit (cyclotomic ring, CRT, NTT, dual lattice) lets Ring-LWE **keep security comparable to LWE while cutting cost from $O(n^2)$ to $O(n \log n)$** .
- The price: security relies on an extra “ideal-lattice” assumption (see open problems).

- **FIPS 203 (ML-KEM = Kyber)**: key encapsulation, based on **Module-LWE** (a generalization of Ring-LWE over module lattices $R_q^{k \times k}$); Kyber-512 uses $n = 256$, $q = 3329$, NIST Level I (attack hardness $\approx 2^{118}$).
- **FIPS 204 (ML-DSA = Dilithium)**: digital signatures, based on Module-LWE + Module-SIS; public key ≈ 1.3 KB, signature ≈ 2.4 KB.
- Module-LWE is a compromise: between Ring-LWE (fastest, most structured) and LWE (most conservative), a small $k \times k$ matrix trades “lower structural risk” for “still NTT-friendly.”

Security chain

breaking the cipher $\iff$ solving BDD on an ideal lattice $\iff$ ideal-SVP $_{\gamma}$ is hard.

- **Ideal-SVP vs. SVP:** is SVP on ideal lattices truly no easier than on arbitrary lattices? Cramer et al. (2016) gave a quantum speedup for certain cyclotomic rings; the general case is open. If ideal-SVP turns out easy, Ring-LWE's security guarantee weakens.
- **Ring-LWE over non-cyclotomic fields:** the LPR reduction relies on the abelian Galois structure of cyclotomic fields; whether it holds for rings whose Galois closure is non-abelian is unknown.
- **Classical worst-case reduction:** Regev's LWE reduction is quantum; a purely classical worst-case $\rightarrow$ average-case reduction has not been found.
- **Quantum lower bound for SVP:** the best quantum algorithm is $2^{0.265n}$ (Laarhoven 2015), yet no unconditional exponential lower bound has been proven.

Thank you

Questions & discussion

O. Regev, *J. ACM* **56**(6), 2009. V. Lyubashevsky–C. Peikert–O. Regev, *J. ACM* **60**(6), 2013.
H. Minkowski, *Geometrie der Zahlen*, 1896. J. Neukirch, *Algebraic Number Theory*, Springer, 1999. NIST
FIPS 203/204, 2024.

Further reading:

<https://hackmd.io/9wQpRveaS4ywg2XBugxF0A>

<https://drive.google.com/file/d/10dtsrpwJK5mDYlakeiCoA-2u0mbF9ydp/view?pli=1>